

Rohan Gupta

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Driven and excited about opportunities in AI/ML (Multimodal, Robotics, Computer Vision, NLP)

EDUCATION

UNIVERSITY OF SOUTHERN CALIFORNIA

BS Computer Science, Finance Minor

MS Computer Science, emphasis in AI/ML & Robotics

GPA: 3.73/4

Relevant Activities: Undergraduate TA for introductory Artificial Intelligence and Operating Systems courses

Awards/Designations: USC Presidential Merit Scholar, Provost's Undergraduate Research Fellowship, Viterbi Dean's List

Los Angeles, CA

Aug 2020 - May 2024

Jan 2024 to May 2025

RELEVANT EXPERIENCE

SYNOPSIS | *Senior Machine Learning Engineer*

R&D Machine Learning Intern

July 2025 – Present

June 2024 – Feb 2025

- Working on a variety of ML projects related to optimizing chip design, using libraries and frameworks like JAX, CUDA, and PyTorch and built featurization pipelines for chip design data, such as netlists and QoR data.
- Conducted experiments in JAX and CUDA to speed up differentiable optimization for both ML and non-ML based methods with chip placement algorithms used in products
- Researched and implemented various architectures, such as Transformers, Diffusion, and others, and adapted them for models related to chip design, such as a foundation model for netlists
- Intern project: Used Python and LangChain to develop multiple LLM-based agents that are set to be incorporated in flagship products like Fusion Compiler, RTL, and ICC2.

USC VITERBI SCHOOL OF ENGINEERING | *AI/ML Research Assistant & Engineer*

May 2023 – July 2025

- Collaborated with Professor Jesse Thomason in the GLAMOR Lab on robotics, computer vision, and NLP research for Vision-and-Language Navigation and Sim2Real
- Collected, trained and fine-tuned language-conditioned policies (ACT, Cross Modal Attention, Diffusion, etc.) on multimodal robot data for mobile manipulators and navigation robots using reinforcement learning for finetuning and supervised learning, specifically behavior-cloning
- Learned ROS1 and ROS2 along a variety of robots' libraries (Toyota HSR, Trossen, Stretch, and some others) and contributed heavily to a working codebase for heterogenous multi-robot coordination project
- Worked on top of public repos like TeleMoMa and LeRobot and fixed bugs in them as well as added ROS1 to ROS2 upgrades for them
- Built a sample-efficient evaluation framework for a physical robot using perplexity and distance-to-goal as success metrics
- Researched how contrast sets can be applied to robot experiments to save time and experimenter effort — co-first author on both papers for this project that got accepted to a [CoRL 2023 Workshop](#) (Oral, Top 15% of papers) and [CoRL 2024](#) (38.2% acceptance rate)
- Current project "RobotFleet" is heterogenous multi-robot coordination using natural language LLMs and multimodal policies. Paper accepted to RSS 2025 Multi-Robot Systems Workshop (not yet public).

USC INVESTMENT OFFICE | *Software Engineering and Data Analyst Intern*

Jun – Dec 2022

- Worked with an 8-person team to help run an \$8+ billion endowment by focusing on software and automation projects.
- Made website on Wordpress and coded a financial news webscraper using Python and BeautifulSoup.
- Performed statistical analysis of financial time-series data using Excel, Python, and other financial software, and took notes at meetings.

SELF-STARTED AND CLASS PROJECTS | *Co-founder, student developer*

- StreetFins*** **Oct 2017 – Nov 2022**
Built and grew team of 60+ students for a financial education startup. Created the business plan, coded the website, and reached out successfully to high schools across 16 states. Hosted the "Finance Simplified" podcast which has over 70k listens with notable guests from industry, academia, and government. Won paid contracts with 2 larger companies to produce educational content.
- Peersdom*** **Oct 2017 – Mar 2019**
Built and led team of 10+ students on app development, using the Ionic Angular framework for the frontend in TypeScript, JavaScript, HTML, CSS, Google Cloud for the SQL-based backend, and Firebase for authentication and messaging.
- Capstone Project (Data Analytics Project for Chris Mattmann, former CTIO of NASA JPL)*** **Jan – May 2023**
Transformed datasets from website analytics tools to reveal insights about visitor data. Coded NLP tools using Python, Pandas, Top2Vec, spaCy, and Tika in addition to heavy Excel work.
- Classical Music Generation*** **Sep – Dec 2023**
Used the music21 library and PyTorch to do supervised learning and train transformers, RNNs, and n-grams to generate classical music. Created a custom data transformation pipeline for musical compositions from 9 classical composers and trained models on that data.

ADDITIONAL INFORMATION

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, Angular, React, C, PHP, SQL
- Frameworks/Tools:** JAX, CUDA, PyTorch, Git, AWS, Google Cloud, CLI, Docker, CI/CD, Linux/Unix, ROS, LangChain, OpenCV, NumPy, Hydra, WandB, Vector Databases